

assignment 12 stellar parameters 7.1 and 7.2, 7.3

 This is a preview of the published version of the quiz

Started: Mar 22 at 7:43pm

Quiz Instructions

If you have any question, email me or stay after class

luminosity depends on the Mass to the 4th power or temperature to the 4th power

life depends on the mass to the 3rd power (inverse relationship). Like if mass x 10 then life is divided by 1000.

Life is inversely proportional to luminosity



Question 1 4 pts

Who was the first woman to get a PhD at Radcliffe College, at the time the women's branch of Harvard University in Cambridge, Massachusetts Harvard in 1925?

☐

Marie Curie

☐

Annie Jump Cannon

☐

Cecilia Payne

☐

jocelyn Bell



Question 2 4 pts

For main-sequence stars, which of the following is *true* about low-mass stars compared to high-mass stars?

☐

Low-mass stars are cooler but more luminous than high-mass stars.

☐

Low-mass stars are hotter and more luminous than high-mass stars.

☐

Low-mass stars have the same temperature and luminosity as high-mass stars.



Low-mass stars are cooler and less luminous than high-mass stars.



Low-mass stars are hotter but less luminous than high-mass stars.



Question 3 4 pts

On a Hertzsprung-Russell diagram, where would we find stars that are cool and luminous?

(see picture in next question)



upper right



lower left



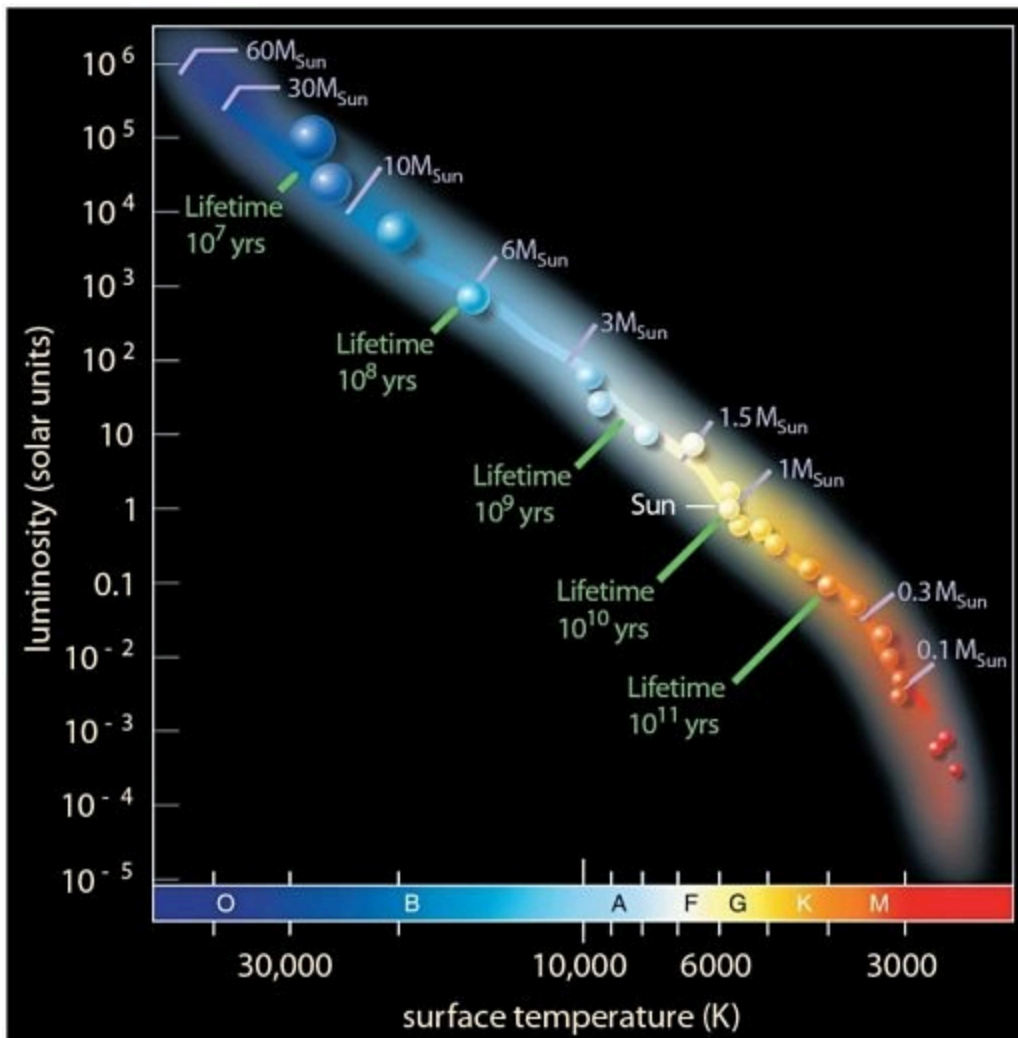
lower right



upper left



Question 4 4 pts



Based on the H-R diagram shown above, what is the approximate surface temperature of an F star?

- ☐ 7000 K
- ☐ 5000 K
- ☐ 6000 K
- ☐ 12,000 K



Question 5 4 pts

Betelgeuse is more luminous than the Sun but cooler because it is

- ☐ larger
- ☐ smaller



running out of fuel



about to die



Question 6 4 pts

Which spectral type represents stars with the hottest surface temperatures?



A



G



B



M



O



Question 7 4 pts

Based on the H-R diagram shown above, what is the approximate surface temperature of a B star?



6000 K



20,000 K



40,000 K



8000 K



Question 8 4 pts

Who (Physicist) made the connection between the spectra of stars (absorption lines) and their temperature.



Max Planck



Maghanad Saha



Wien



Annie Jump



Question 9 4 pts

What physical property of a star does the spectral type (like A type) measure? (see slide 22 - stellar parameters1)

☐

mass

☐

luminosity

☐

density

☐

composition

☐

temperature



Question 10 4 pts

The lumosity is the intrinsec power of a star. If the distance from the star is multiplied by 3, its brightness is divided by (see my slide about inverse square law. slide 2 stellar parameter)

☐

2

☐

16

☐

8

☐

9



Question 11 4 pts

Which of the following best describes the axes of a Hertzsprung-Russell (H-R) diagram?

☐

surface temperature on the horizontal axis and luminosity on the vertical axis

☐

surface temperature on the horizontal axis and radius on the vertical axis

☐

mass on the horizontal axis and stellar age on the vertical axis

☐

interior temperature on the horizontal axis and mass on the vertical axis

☐

mass on the horizontal axis and luminosity on the vertical axis



Question 12 4 pts

Consider a star 5 times more massive than the Sun. It will be _____ times more luminous than the Sun. (L depends on mass Mass to the 4th power)

☐

5

☐

25

☐

625

☐

125

☒

Question 13 4 pts



Alnitak is a O star. It's about 20 times the mass of the Sun. What is life expectancy (Sun is 10 billion)

hint: 10 billion = 10,000,000,000 . The life is inversely proportional to the mass to the third power. So divide by $20 \times 20 \times 20$

☐

about 1 billion

☐

about 1 million

☐

about 10,000

☐

about 10 million

☒

Question 14 4 pts

Which of the following people reorganized the spectral classification scheme into the one we use today and personally classified over 400,000 stars? hint: that person came up with

"oh boy a fine ... "

☐

Cecilia Payne-Gaposchkin

☐

Annie Jump Cannon



Williamina Fleming



Edward Pickering



Henry Draper



Question 15 4 pts

What is a star's *luminosity*?



the apparent brightness of the star in our sky



the total energy that the star radiates each second



the lifetime of the star



the surface temperature of the star



Question 16 4 pts

The Stefan-Boltzmann laws (slide 14 unit 7.2) gives the relationship between ... and the temperature of a star



There is not relationship between temperature and energy output



energy output per second and per unit area



energy output per unit second



energy per unit area



Question 17 4 pts

A 1000 watts light bulb placed at twice the distance from you will appear (inverse square law)



100W



500W



250W



1000W



Question 18 4 pts

Suppose that the Sun—Earth distance were 10 AU instead of 1 AU. Then the energy of sunlight received per square meter on Earth each second would be _____ it is today.

(so distance is x 10 so the luminosity is divided by ? Its an onverse square law slide 2)

☐

about the same as

☐

1/100 as much as

☐

10 times as great as

☐

1/10 as much as

☐

100 times as great as



Question 19 4 pts

Henrietta Leavitt Studied cepheid variables in

☐

Andromeda galaxy

☐

eagle nebula

☐

magellanic clouds

☐

magestic clouds



Question 20 4 pts

The absolute magnitude measures the same thing as

☐

mass

☐

luminosity

☐

size

☐

brightness



Question 21 4 pts

Suppose the distance to a star was doubled but everything else about the star stayed the same. What would happen to the star's luminosity and apparent brightness? (hint: inverse square law)

☐

Luminosity is decreased by a factor of two and the apparent brightness is decreased by a factor of two.



Luminosame, but the apparent brightness is decreased by a factor of four.ity remains the s



Luminosity remains the same, but the apparent brightness is decreased by a factor of two.



Luminosity is decreased by a factor of four and the apparent brightness is decreased by a factor of four.



Luminosity is decreased by a factor of four, but the apparent brightness remains the same.



Question 22 4 pts



R136a1 is a really hot and big star. Here it was compared to the Sun. Its mass was 300 times the mass of the Sun! Say 100 times more mass. So its life is about?

(hint: divide life of sun (10 billion) by 100 raised to the third power).



100,000 years



10,000 years



1 million years



10 million years



Question 23 4 pts

Who classifies stars according to their apparent magnitude



Max Plank



Annie Jump Cannon



Hipparchus



Cecilia Payne



Question 24 4 pts

O stars don't have many H absorption lines because



hydrogen electrons don't like the color blue



There is no hydrogen in O star



Its so hot that all hydrogen electrons are free



Its so cool that hydrogen electrons can't make their leap to higher energy level



Question 25 4 pts

The stellar classes are



M, L, T, O, G, K, A, B, F



O, B, A, F, G, T, L, M, K



G, K, M, L, T, O, B, A, F,



O, B, A, F, G, K, M, L, T



Question 26 4 pts

Henrietta Leavitt found a relationship between



luminosity of cepheids and their period



luminosity of cepheids and their size



peak wavelength of cepheids and their temperature



luminosity of cepheids and their temperature



Question 27 4 pts

To get the absolute magnitude of stars, place them at a distance of about.. from Earth

(see unit 7.2 about absolute magnitude)

☐

200 L. Y

☐

30 L. Y

☐

1 L. Y

☐

3 L.Y

Not saved

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